main.py > ...

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

np.random.seed(42)

n = 1000

data = pd.DataFrame({
    'Visitor_ID': range(1, n+1),
    'Page_Views': np.random.randint(1, 15, n),
    'Session_Duration_Min': np.random.randint(1, 30, n),
    'Bounce_Rate': np.random.uniform(20, 90, n),
    'Traffic_Source': np.random.choice(
        ['Organic Search', 'Direct', 'Social Media', 'Referral'],
        n
    ),
    'Device': np.random.choice(
        ['Desktop', 'Mobile', 'Tablet'],
        n
    ),
    'Conversions': np.random.choice(
        [0,1],
        n,
        p=[0.8,0.2]
    ),
    'Revenue': np.random.randint(0, 5000, n)
})

print("Dataset Shape:", data.shape)
print("\nFirst 5 Records")
print(data.head())

print("\nSummary Statistics")
print(data.describe())
```

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```python
plt.figure(figsize=(8,5))
sns.countplot(
    data=data,
    x='Traffic_Source',
    palette='Set2'
)
plt.title("Visitors by Traffic Source")
plt.xticks(rotation=20)
plt.show()

device_counts = data['Device'].value_counts()

plt.figure(figsize=(7,7))
plt.pie(
    device_counts,
    labels=device_counts.index,
    autopct='%1.1f%%'
)
plt.title("Device Distribution")
plt.show()

plt.figure(figsize=(8,5))
sns.histplot(
    data['Page_Views'],
    bins=15,
    kde=True
)
plt.title("Page Views Distribution")
plt.show()

plt.figure(figsize=(8,5))
sns.scatterplot(
    data=data,
    x='Session_Duration_Min',
    y='Revenue',
    hue='Conversions'
)
```

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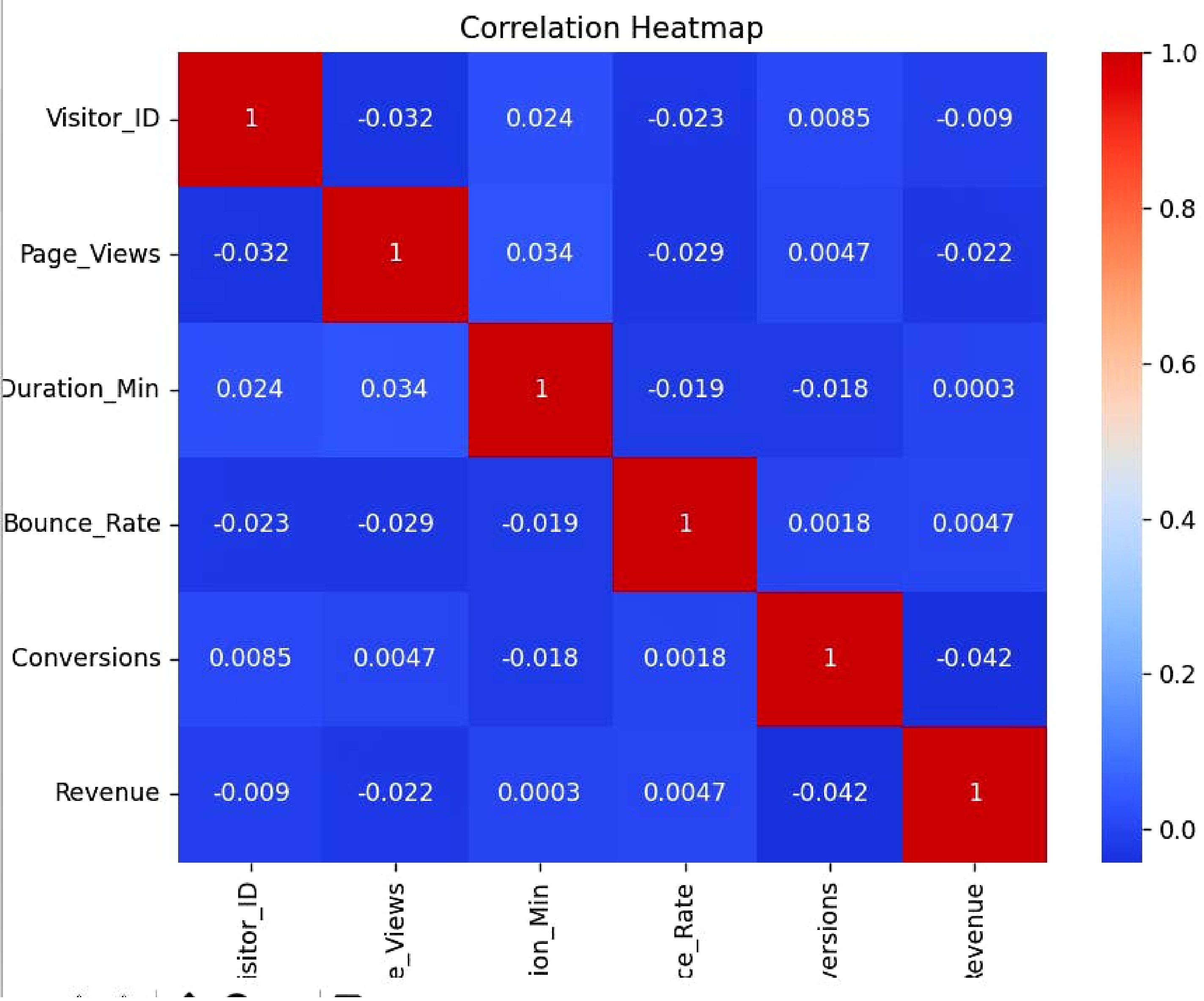
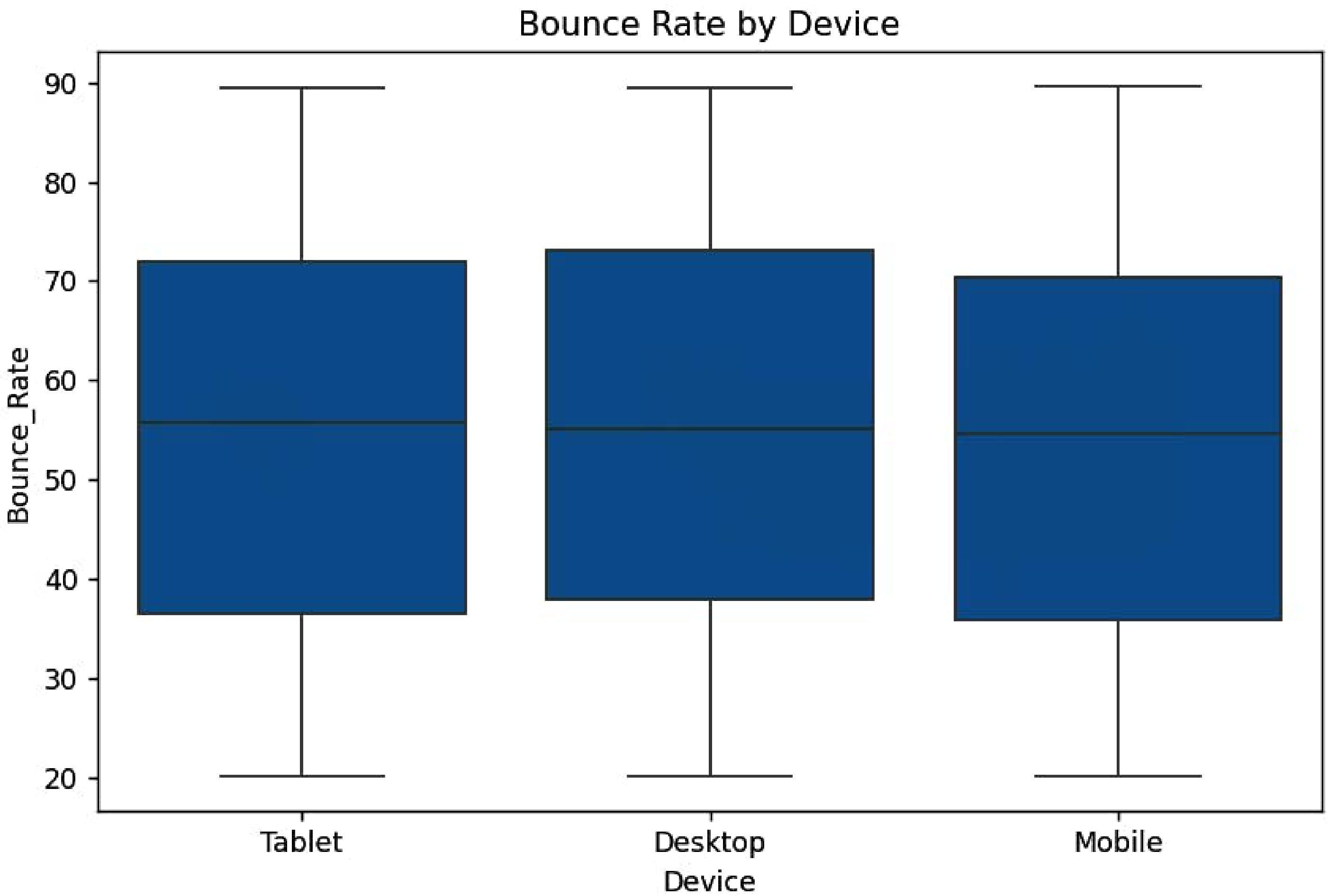
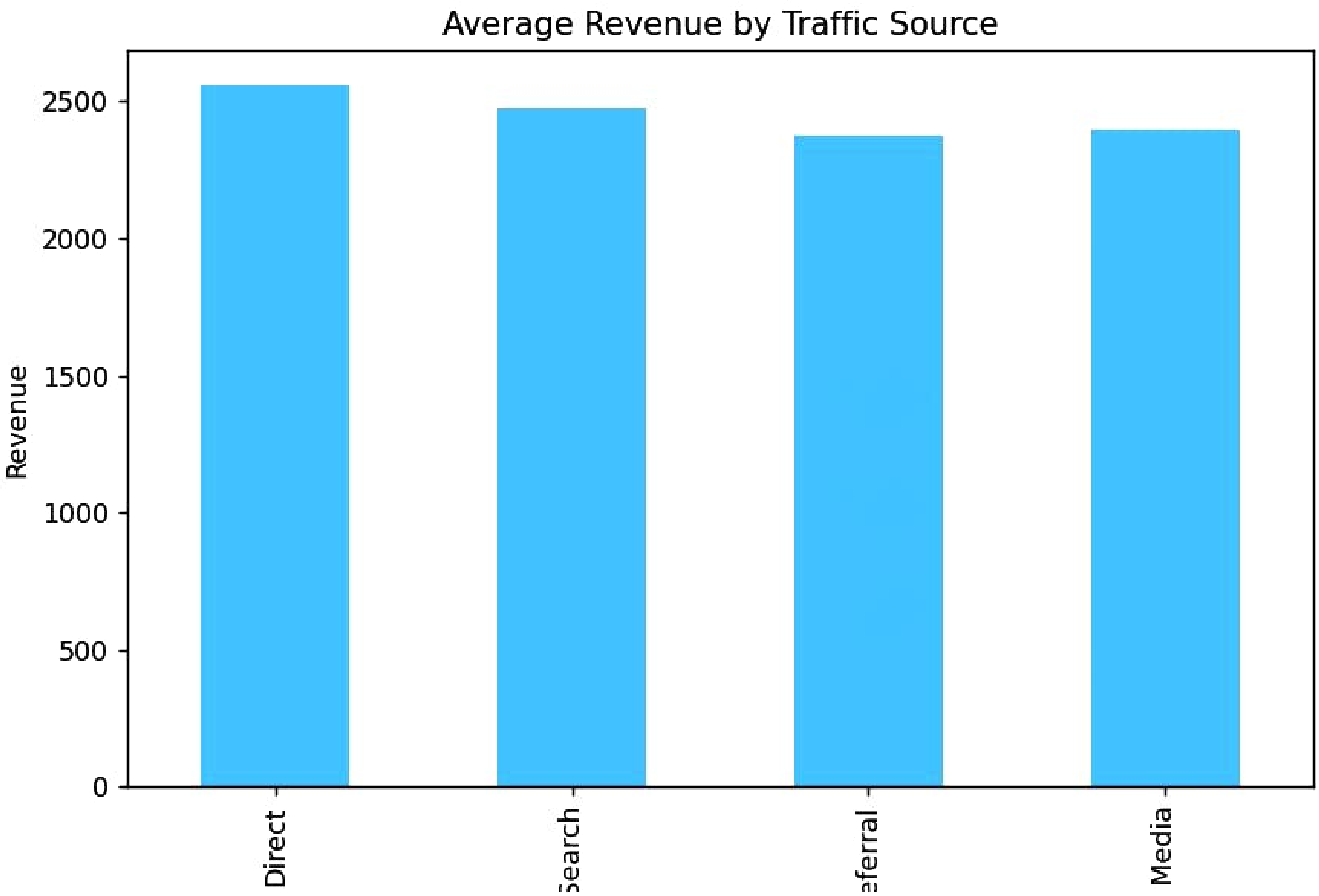
```python
plt.title("Session Duration vs Revenue")
plt.show()

conversion_rate = (
    data['Conversions'].mean()*100
)

print("\nOverall Conversion Rate:")
print(round(conversion_rate,2), "%")

source_revenue = data.groupby(
    'Traffic_Source'
)['Revenue'].mean()

print("\nAverage Revenue by Source")
print(source_revenue)

plt.figure(figsize=(8,5))
source_revenue.plot(
    kind='bar',
    color='skyblue'
)
plt.title("Average Revenue by Traffic Source")
plt.ylabel("Revenue")
plt.show()

plt.figure(figsize=(8,5))
sns.boxplot(
    data=data,
    x='Device',
    y='Bounce_Rate'
)
plt.title("Bounce Rate by Device")
plt.show()

numeric_cols = data.select_dtypes(
    include=np.number
```

```python
plt.title("Bounce Rate by Device")
plt.show()

numeric_cols = data.select_dtypes(
    include=np.number
)

plt.figure(figsize=(8,6))
sns.heatmap(
    numeric_cols.corr(),
    annot=True,
    cmap='coolwarm'
)
plt.title("Correlation Heatmap")
plt.show()

print("\nWebsite Analytics Completed Successfully")
```

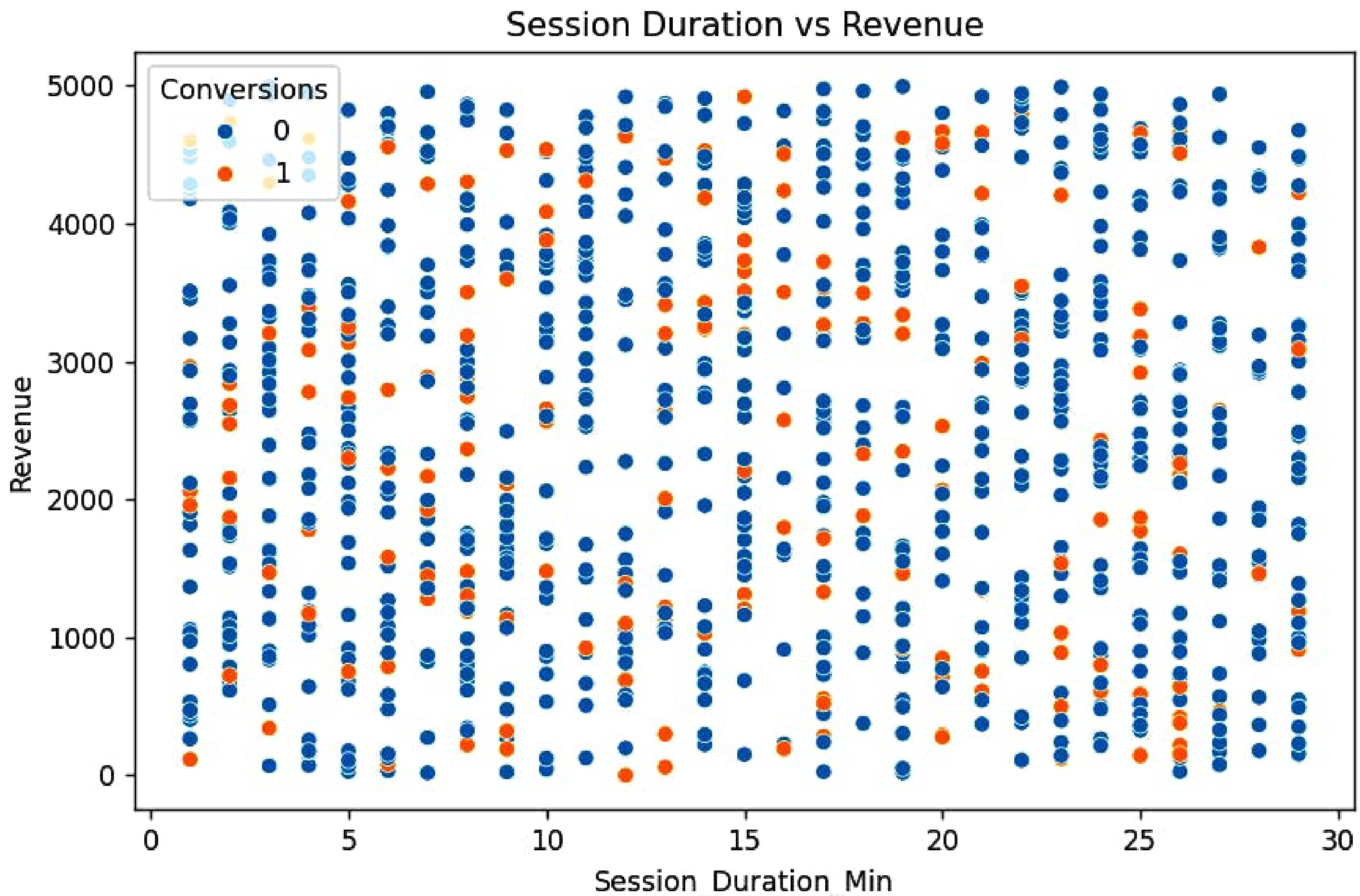
```
Dataset Shape: (1000, 8)

First 5 Records
   Visitor_ID  Page_Views  Session_Duration_Min  Bounce_Rate  Traffic_Source   Device  Conversions  Revenue
0           1           7                    19    79.653547    Social Media   Tablet            0     1211
1           2           4                    17    22.479566        Referral  Desktop            0      929
2           3          13                     5    43.033596  Organic Search   Tablet            0     1539
3           4          11                    29    85.211264    Social Media  Desktop            1      912
4           5           8                     4    71.464886  Organic Search   Mobile            0      643

Summary Statistics
        Visitor_ID   Page_Views  Session_Duration_Min  Bounce_Rate  Conversions      Revenue
count  1000.000000  1000.000000           1000.000000  1000.000000  1000.000000  1000.000000
mean    500.500000     7.443000             14.962000    54.625716     0.195000  2449.447000
std     288.819436     4.113596              8.419944    19.975657     0.396399  1430.517413
min       1.000000     1.000000              1.000000    20.023275     0.000000     1.000000
25%     250.750000     4.000000              8.000000    36.558684     0.000000  1188.500000
50%     500.500000     7.000000             15.000000    55.241328     0.000000  2468.000000
75%     750.250000    11.000000             23.000000    72.126799     0.000000  3653.500000
max    1000.000000    14.000000             29.000000    89.683634     1.000000  4994.000000
C:\Users\DHARSHINI P\OneDrive\Desktop\main.py:39: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hu
e` and set `legend=False` for the same effect.

  sns.countplot(

Overall Conversion Rate:
19.5 %

Average Revenue by Source
Traffic_Source
Direct            2559.056277
Organic Search    2473.764259
Referral          2374.314286
Social Media      2398.459770
Name: Revenue, dtype: float64

Website Analytics Completed Successfully
```


Correlation Heatmap
Visitor_ID
Page_Views
Duration_Min
Bounce_Rate
Conversions
Revenue
1
-0.032
0.024
-0.023
0.0085
-0.009
-0.032
1
0.034
-0.029
0.0047
-0.022
0.024
0.034
1
-0.019
-0.018
0.0003
-0.023
-0.029
-0.019
1
0.0018
0.0047
0.0085
0.0047
-0.018
0.0018
1
-0.042
-0.009
-0.022
0.0003
0.0047
-0.042
1
1.0
0.8
0.6
0.4
0.2
0.0

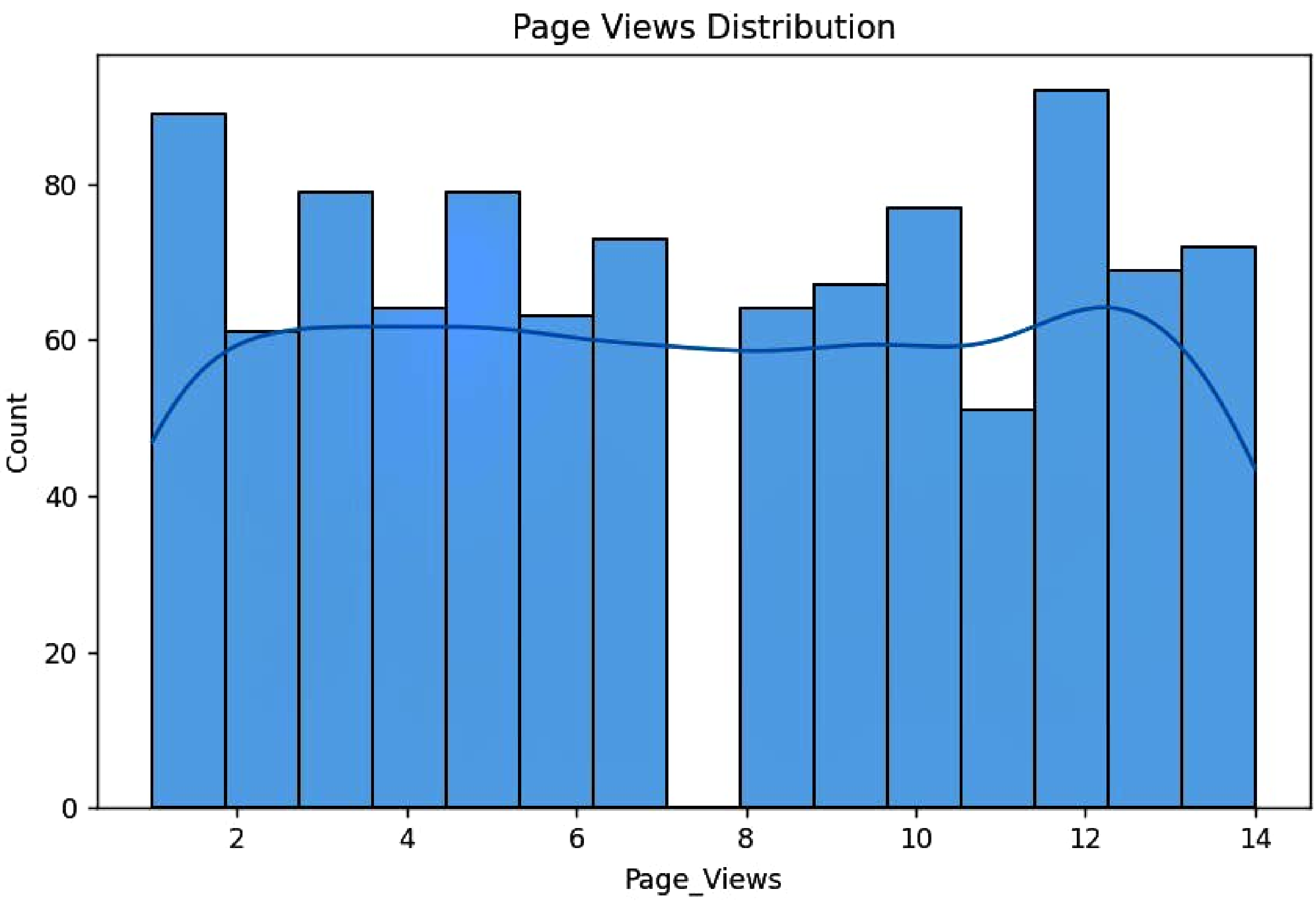

Bounce Rate by Device
Bounce_Rate
90
80
70
60
50
40
30
20
Tablet
Desktop
Mobile
Device



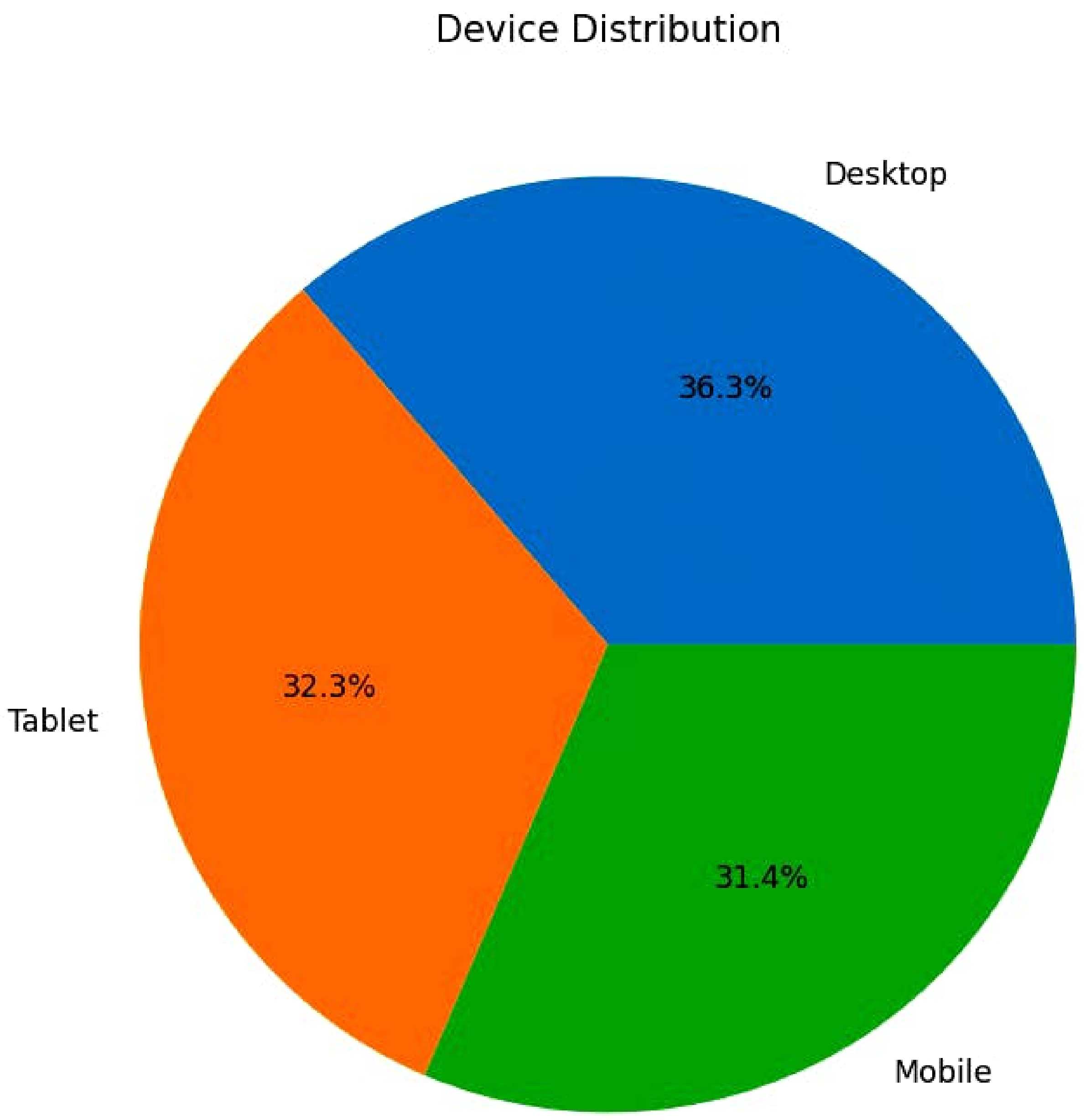
Average Revenue by Traffic Source
Revenue
2500
2000
1500
1000
500
0
Direct
Search
eferral
Media

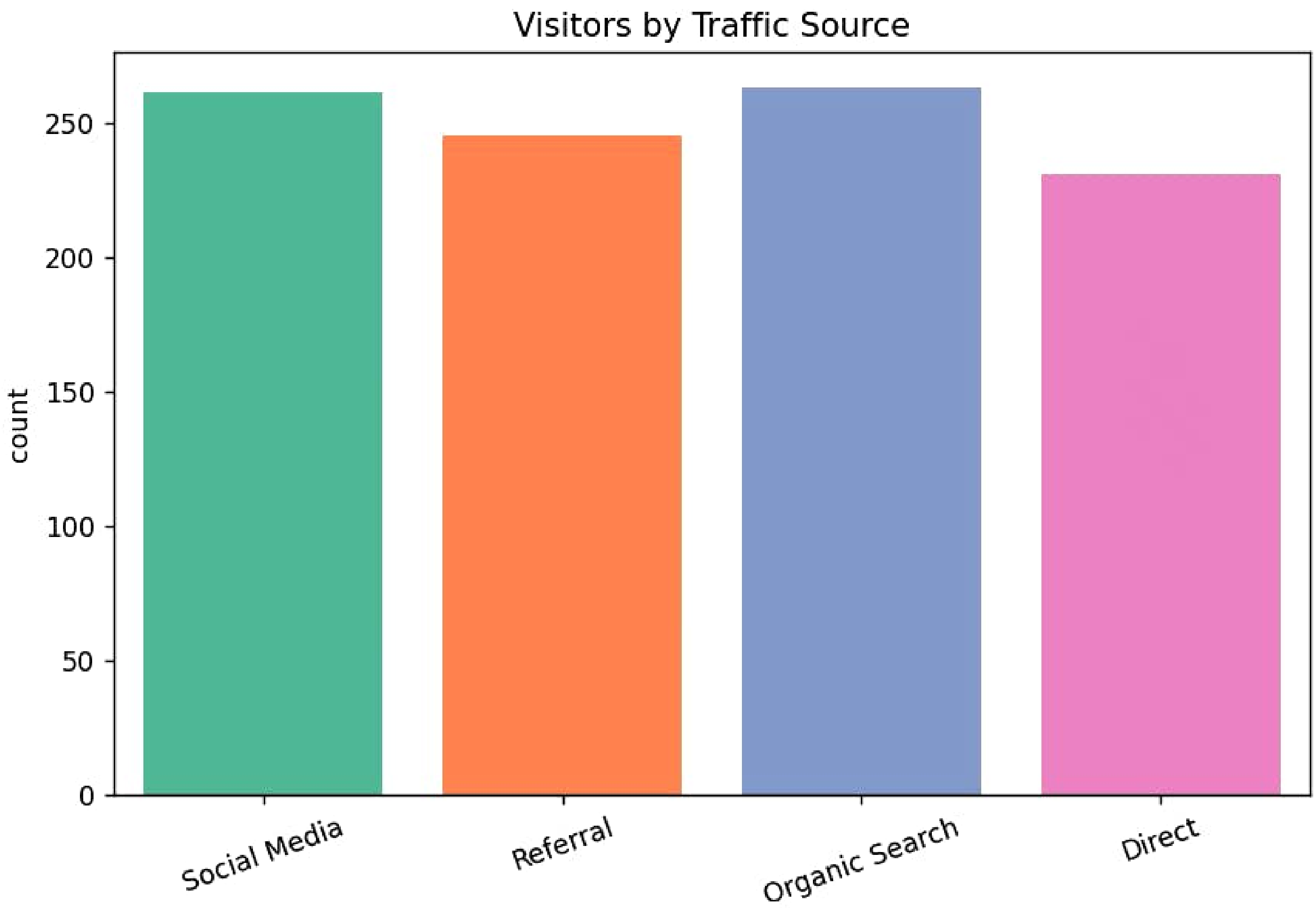


Session Duration vs Revenue
Conversions
0
1
Revenue
Session_Duration_Min
0
1000
2000
3000
4000
5000
0
5
10
15
20
25
30



Page Views Distribution
Count
Page_Views
0
20
40
60
80
2
4
6
8
10
12
14


Device Distribution
Desktop
36.3%
Tablet
32.3%
Mobile
31.4%


Visitors by Traffic Source
count
0
50
100
150
200
250
Social Media
Referral
Organic Search
Direct